

SULIT



SECP2523 DATABASE

SESSION 2024/2025 - SEMESTER 1

ALTERNATIVE ASSESSMENT REPORT: PHASE 2

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Case Study	Koperasi Kakitangan KADA (KKK)
System Name	KKK Online System
Group Name	Tech Hi-Five

SECTION C (SQL AND INTERFACE IMPLEMENTATION)

3.1 SQL Statement (DDL & DML)

DDL

tb_chat

```
1 CREATE TABLE `tb_chat` (  
2   `c_id` int(11) NOT NULL,  
3   `c_sender` int(11) NOT NULL,  
4   `c_receiver` int(11) NOT NULL,  
5   `c_msg` varchar(100) NOT NULL,  
6   `c_time` timestamp NOT NULL DEFAULT current_timestamp()  
7 ON UPDATE current_timestamp()  
8 )  
9  
1  -- Indexes for table `tb_chat`  
0  --  
1  ALTER TABLE `tb_chat`  
1    ADD PRIMARY KEY (`c_id`);  
1  
2  -- AUTO_INCREMENT for table `tb_chat`  
1  --  
3  ALTER TABLE `tb_chat`  
1    MODIFY `c_id` int(11) NOT NULL AUTO_INCREMENT;  
4  
1  ADD CONSTRAINT `tb_chat_ibfk_1` FOREIGN KEY (`c_sender`)  
5  REFERENCES `tb_member` (`m_memberNo`)  
1  
6  
1  
7  
1  
8  
1  
9  
2  
0  
2  
1
```

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0008 seconds.)

SELECT * FROM `tb\_chat`

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

c_id c_sender c_receiver c_msg c_time

tb_inquiries

```
1 CREATE TABLE `tb_inquiries` (  
2   `q_id` int(11) NOT NULL,
```

```

3      `q_question` varchar(1000) NOT NULL,
4      `q_answer` varchar(1000) NOT NULL
5  )
6
7  -- Dumping data for table `tb_inquiries`
8  --
9
10 INSERT INTO `tb_inquiries` (`q_id`, `q_question`,
11 `q_answer`) VALUES
12 (1, 'Bagaimanakah cara untuk mengemaskini maklumat
13 saya?', 'Tekan pada profil, kemudian pilih kemas kini
14 maklumat dan buat perubahan yang diperlukan. Apabila
15 selesai, tekan simpan.'),
16 (2, 'Apakah syarat minimum untuk memohon pinjaman?',
17 'Syarat minimum ialah pendapatan bulanan mesti melebihi
18 RM1000. (contoh)'),
19 (3, 'Jika saya menarik diri, bolehkah saya menjadi ahli
20 semula?', 'Boleh, tetapi terdapat syarat tertentu.');
```

-- Indexes for table `tb\_inquiries`

```

21 --
22 ALTER TABLE `tb_inquiries`
23   ADD PRIMARY KEY (`q_id`);
24
25 ALTER TABLE `tb_inquiries`
26   MODIFY `q_id` int(11) NOT NULL AUTO_INCREMENT;
```

✓ Showing rows 0 - 2 (3 total, Query took 0.0006 seconds.)

`SELECT * FROM `tb_inquiries``

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	q_id	q_question	q_answer
☐ Edit Copy Delete	1	Bagaimanakah cara untuk mengemaskini maklumat saya...	Tekan pada profil, kemudian pilih kemas kini maklu...
☐ Edit Copy Delete	2	Apakah syarat minimum untuk memohon pinjaman?	Syarat minimum ialah pendapatan bulanan mesti mele...
☐ Edit Copy Delete	3	Jika saya menarik diri, bolehkah saya menjadi ahli...	Boleh, tetapi terdapat syarat tertentu.

DML

Use Case: Chat with Chatbot

```

1 // Fetch all predefined questions from the database
2 $sql = "SELECT q_id, q_question FROM tb_inquiries";
3 $result = $con->query($sql);
4
5
6 // Get answer at get answer.php by question id
7 $q_id = isset($_GET['q_id']) ? intval($_GET['q_id']) : 0;
8
9
10 if ($q_id > 0) {
11     try {
12         $stmt = $con->prepare("SELECT q_answer FROM
13 tb_inquiries WHERE q_id = ?");
14         $stmt->bind_param("i", $q_id);
15         $stmt->execute();
16         $result = $stmt->get_result();
17     }
18 }

```

Use Case: Chat with Member (Admin)

```

1 //receive message
2 $sqlReceiveMessages = " SELECT c_msg AS message, c_time
3 AS time, c_sender AS sender FROM tb_chat WHERE c_receiver
4 = '$sessionId'; ";
5
6 //send message
7
8

```

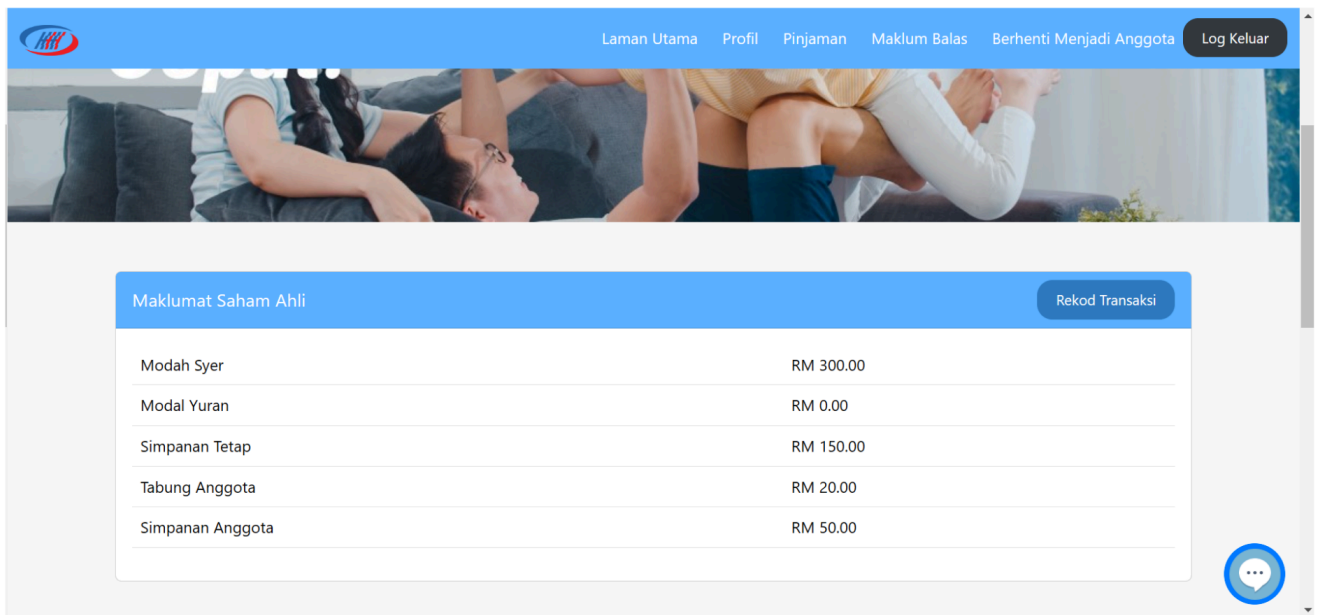
```
$sqlSendMessage = " INSERT INTO tb_chat (c_sender,  
c_receiver, c_message, c_time) VALUES ('$sessionId',  
'$msgSender', '$message', '$current_time'); ";
```

Use Case: Chat with Admin (Member)

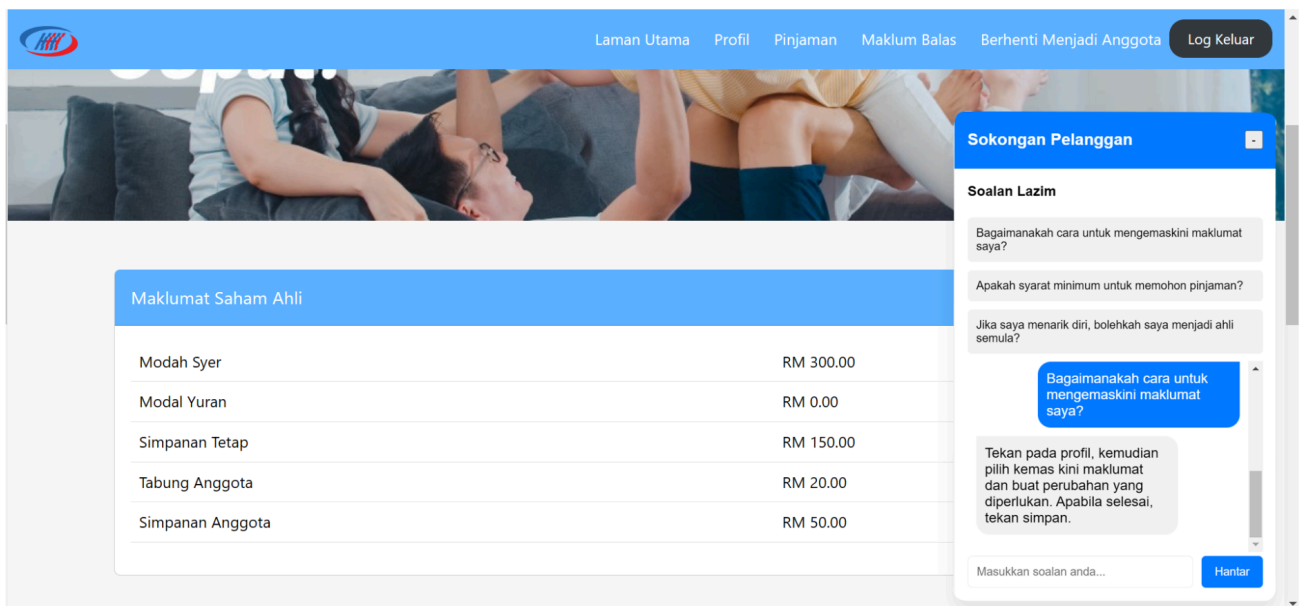
```
1 //check for all admin  
2 $sqlGetAdmins = " SELECT u_id FROM tb_user WHERE u_type =  
3 1;  
4  
5 //receives message from admin  
6 $sqlReceiveMessages = " SELECT c_sender AS adminId,  
7 c_message AS message, c_time AS time FROM tb_chat WHERE  
8 c_sender = '$adminId' AND c_receiver = '$sessionId'; ";  
9  
10 //select one admin  
11 $sqlGetOneAdmin = " SELECT u_id FROM tb_user WHERE u_type  
12 = 1 LIMIT 1; ";  
13  
14 //send message  
15 $sqlSendMessage = " INSERT INTO tb_chat (c_sender,  
16 c_receiver, c_message, c_time) VALUES ('$sessionId',  
17 '$receiver', '$message', '$current_time'); ";  
18  
19  
20  
21  
22  
23  
24  
25  
26
```

3.2 Interface Implementation

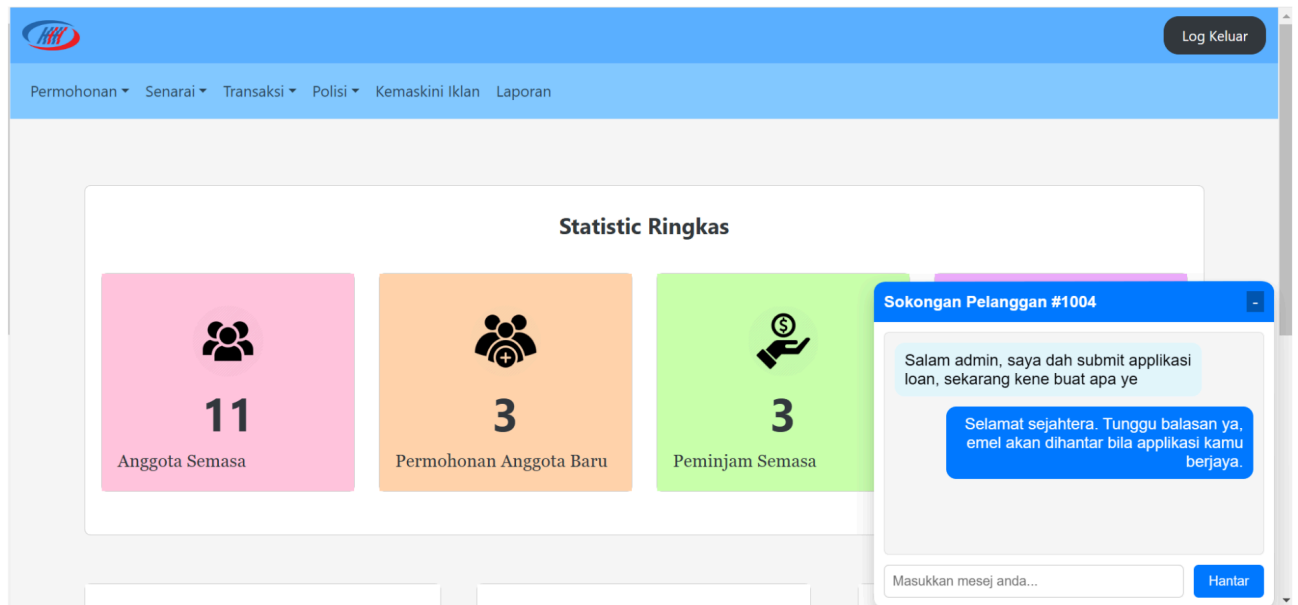
Customer Support Chatbot Interface (Minimised)



Customer Support Chatbot Interface (When clicked)



Messaging Interface (admin)



SECTION D (CONCLUSION AND REFLECTION)

4.1 Conclusion

In conclusion, this module has successfully overcome the problem of lack of communication between member and admin of the KKK system. Member are able to send messages and get admin reply, view past messages, view questions and answers from chatbot. For admin, they can see member messages as an individual window and can reply to their questions.

In the future, it is good to implement more of the live chat functionalities including show online status, online or offline and message status, whether the message is delivered, received or seen. These addition can make communication easier. Furthermore, using the offline/online logic, an admin function of view member online status can also be implemented.

4.2 Reflection

For me, the WBL project is something our team put all of our might to create. It seems like an impossible thing at first, but everything is slower coming into places after endless night of struggling. We did the project without relying on any framework, which is quite a challenge, but that's where I really started to learn about the basics of PHP based web system. In the system design phase last year, I had a lot of idea of what a great system should look like, and that is what we try to achieve in the KKK System to really make it easy to use for every target user.

Through the project, what I learnt about database was creating database, managing the tables such as a row of data should not contain a column with multiple values, and that we have primary key and foreign key which is often referenced between tables, and the logic behind SQL statements to achieve what we want to retrieve/insert, which seems like a mess at first. I feel that something like these are things that are so basic, yet so important for a rookie programmer like me to master.

My regrets in this project is I learned everything too slowly, too struggle, imagine if I was more capable, I would have so many functions to add and make it even better. Even so, I am proud of it as it is truly something our team made with hard work, and in fact the first ever complete system that we have ever made.